

Patient 14 Raw IMC Data Visualization

05-27-2026

Contents

Overview	1
1. Load Libraries	1
2. Read All TXT Files and Combine	1
3. Pivot to Long Format	2
4. Histograms: All Channels, One ROI	2
5. Density Plots: Selected Markers Across All ROIs	4
6. Density Overlay: All Channels for One ROI	6
7. Summary Table: Per-Channel Median and % Zero Pixels	6

Overview

This notebook visualizes pixel-level ion count distributions from raw IMC TXT files — **before any steinbock preprocessing**. Each TXT file is one ROI; rows are pixels, columns are already labeled with both protein name and metal tag (e.g. SMA(Pr141Di)), so no external panel file is needed.

1. Load Libraries

```
library(tidyverse)
library(ggribes)

dir.create("images", showWarnings = FALSE)
```

2. Read All TXT Files and Combine

TXT files live in the raw data directory, one per ROI. Column names already contain the protein name and metal tag (e.g. SMA(Pr141Di)), so no renaming is needed. We drop acquisition metadata and non-biological channels, apply arcsinh transformation (cofactor = 1), and stack everything into a single long-format tibble.

```
cofactor <- 1

txt_files <- list.files(
  path      = "/restricted/projectnb/camplab/projects/20220504_Suzuki_LN/Data/Imaging/IMC/Tumor/Pardee",
  pattern   = "\\..txt$",
  recursive = TRUE,
  full.names = TRUE
)

# acquisition metadata and non-biological channels to drop
exclude_cols <- c("Start_push", "End_push", "Pushes_duration",
                 "X", "Y", "Z",
```

```

      "DNA191(Ir191Di)", "DNA193(Ir193Di)", "Membrane(Pt196Di)")

raw_pixels <- map_dfr(txt_files, function(f) {
  df <- read_tsv(f, show_col_types = FALSE)

  # downsample to 50k pixels per ROI before any heavy operations
  df <- df %>% slice_sample(n = 50000)

  # drop metadata and non-biological channels
  df <- df %>% select(-any_of(exclude_cols))

  # arcsinh transform all remaining numeric columns
  df <- df %>% mutate(across(where(is.numeric), ~asinh(. / cofactor)))

  # tag with ROI label and patient ID parsed from filename
  fname      <- tools::file_path_sans_ext(basename(f))
  df$roi      <- fname
  df$patient <- str_extract(fname, "^[_]+") # everything before first underscore
  df
})

cat("Total pixels loaded:", nrow(raw_pixels), "\n")

## Total pixels loaded: 200000

cat("Patients found:", n_distinct(raw_pixels$patient), "\n")

## Patients found: 1

cat("ROIs found:", n_distinct(raw_pixels$roi), "\n")

## ROIs found: 4

```

3. Pivot to Long Format

```

plot_long <- raw_pixels %>%
  pivot_longer(
    cols      = -c(roi, patient),
    names_to   = "marker",
    values_to  = "intensity"
  )

single_roi <- unique(plot_long$roi)[1]

```

4. Histograms: All Channels, One ROI

A quick per-channel histogram faceted view for a single ROI to get an overview of the intensity landscape.

```

single_roi <- unique(plot_long$roi)[1]

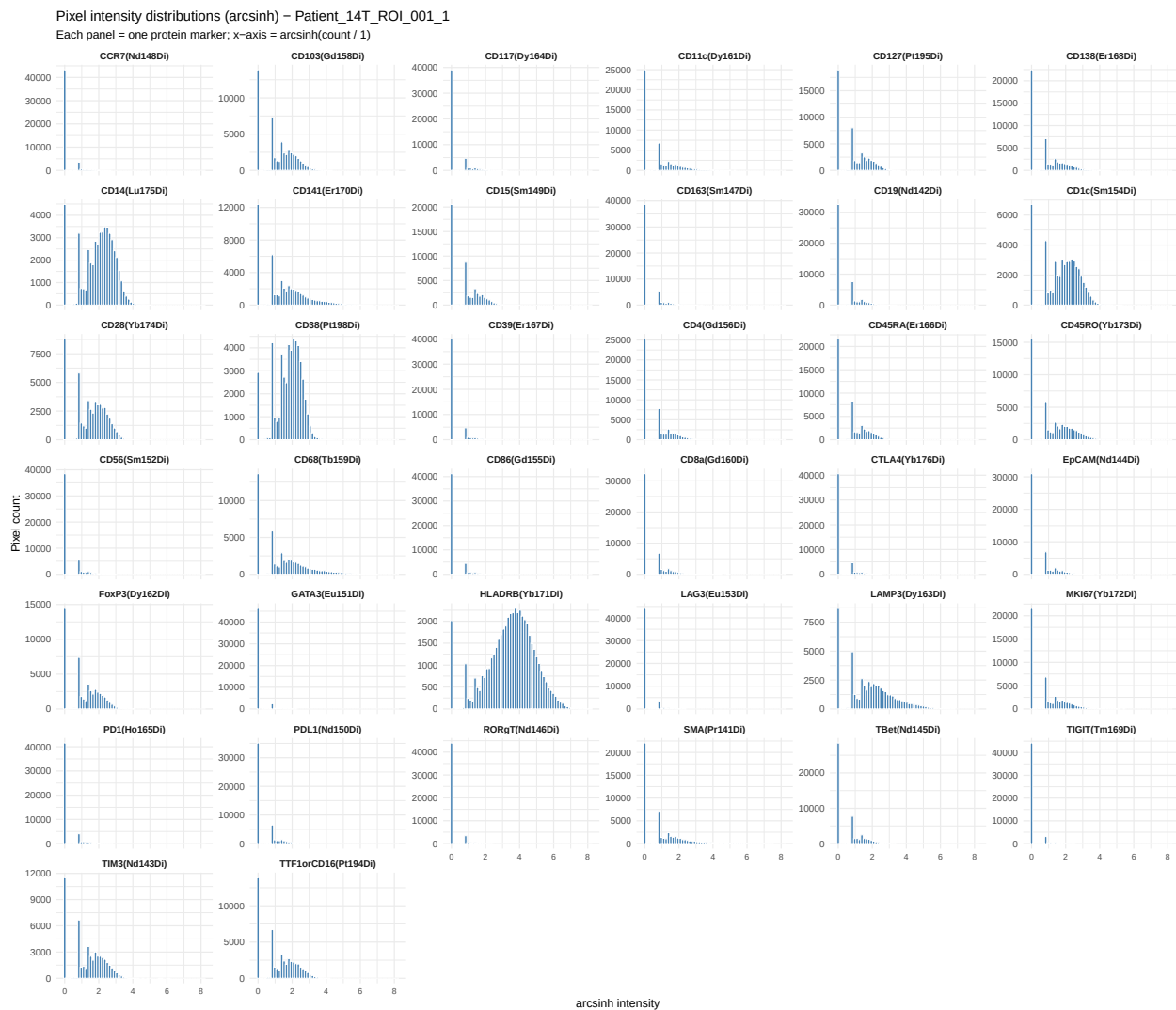
plot_long %>%
  filter(roi == single_roi) %>%

```

```

ggplot(aes(x = intensity)) +
  geom_histogram(bins = 60, fill = "steelblue", color = "white", linewidth = 0.2) +
  facet_wrap(~marker, scales = "free_y", ncol = 6) +
  labs(
    title = paste0("Pixel intensity distributions (arcsinh) - ", single_roi),
    subtitle = "Each panel = one protein marker; x-axis = arcsinh(count / 1)",
    x = "arcsinh intensity",
    y = "Pixel count"
  ) +
  theme_minimal(base_size = 10) +
  theme(
    strip.text = element_text(face = "bold", size = 8),
    axis.text.x = element_text(size = 7)
  )

```



5. Density Plots: Selected Markers Across All ROIs

Ridge plots let you compare intensity distributions across ROIs for a few markers of interest, which is useful for spotting between-ROI batch effects or acquisition issues.

```
p <- plot_long %>%  
  ggplot(aes(x = intensity, y = roi, fill = patient)) +  
  geom_density_ridges(alpha = 0.6, scale = 1.2, rel_min_height = 0.01) +  
  facet_wrap(~marker, scales = "free_x", ncol = 4) +  
  labs(  
    title = "Per-ROI pixel intensity distributions (all markers)",  
    subtitle = "arcsinh(count / 1); each ridge = one ROI, color = patient",  
    x = "arcsinh intensity",  
    y = NULL,  
    fill = "Patient"  
  ) +  
  theme_ridges(font_size = 9) +  
  theme(strip.text = element_text(face = "bold", size = 7))
```

p

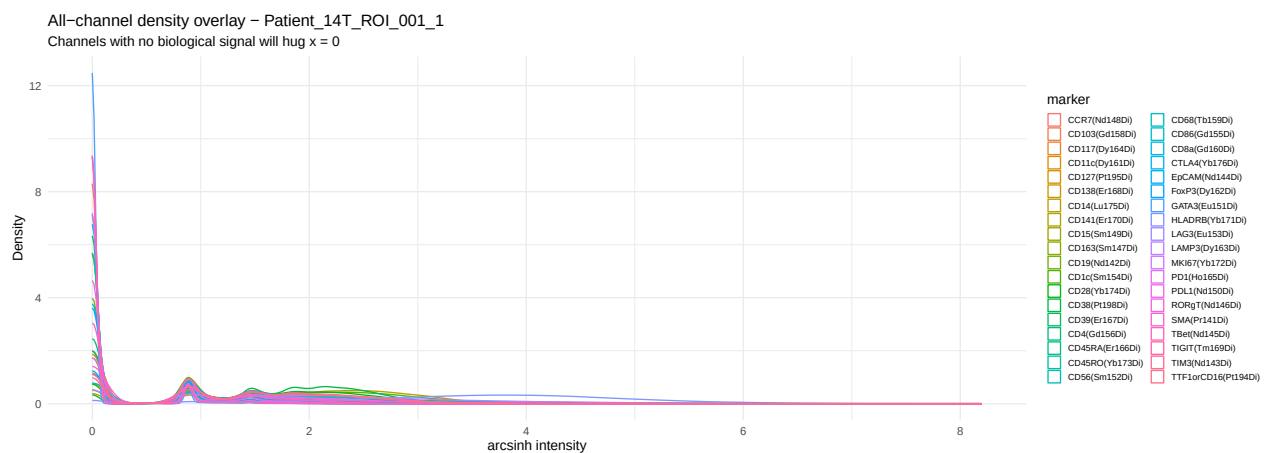


```
ggsave("images/02_ridgeplot_all_markers.png", plot = p, width = 14, height = 40, dpi = 150)
```

6. Density Overlay: All Channels for One ROI

An alternative view: all markers overlaid in a single density plot to compare the signal spread across channels. Useful for spotting channels with no signal or unusually high background.

```
plot_long %>%
  filter(roi == single_roi) %>%
  ggplot(aes(x = intensity, color = marker)) +
  geom_density(linewidth = 0.5, alpha = 0.8) +
  labs(
    title = paste0("All-channel density overlay - ", single_roi),
    subtitle = "Channels with no biological signal will hug x = 0",
    x = "arcsinh intensity",
    y = "Density"
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "right",
        legend.text = element_text(size = 7),
        legend.key.size = unit(0.4, "cm"))
```



7. Summary Table: Per-Channel Median and % Zero Pixels

A numeric summary to flag channels with very low signal (high % zero pixels) or unexpected medians — useful for early QC before committing to preprocessing.

```
plot_long %>%
  group_by(marker) %>%
  summarise(
    median_intensity = round(median(intensity, na.rm = TRUE), 3),
    pct_zero         = round(mean(intensity == 0, na.rm = TRUE) * 100, 1),
    max_intensity    = round(max(intensity, na.rm = TRUE), 2),
    .groups = "drop"
  ) %>%
  arrange(desc(pct_zero)) %>%
  knitr::kable(caption = "Per-channel pixel intensity summary (arcsinh-transformed)")
```

Table 1: Per-channel pixel intensity summary (arcsinh-transformed)

marker	median_intensity	pct_zero	max_intensity
GATA3(Eu151Di)	0.000	91.5	3.10
LAG3(Eu153Di)	0.000	87.6	3.70
TIGIT(Tm169Di)	0.000	86.8	3.23
RORgT(Nd146Di)	0.000	86.6	4.20
CCR7(Nd148Di)	0.000	85.2	6.59
PD1(Ho165Di)	0.000	81.4	3.31
CD86(Gd155Di)	0.000	79.5	5.24
CTLA4(Yb176Di)	0.000	78.0	3.91
CD39(Er167Di)	0.000	77.5	4.00
CD163(Sm147Di)	0.000	75.3	4.32
CD56(Sm152Di)	0.000	74.7	3.25
CD117(Dy164Di)	0.000	74.2	3.96
CD8a(Gd160Di)	0.000	66.7	5.10
PDL1(Nd150Di)	0.000	66.6	3.95
CD19(Nd142Di)	0.000	58.2	5.13
EpCAM(Nd144Di)	0.000	58.2	3.98
TBet(Nd145Di)	0.000	56.4	4.62
CD11c(Dy161Di)	0.881	46.6	5.31
CD4(Gd156Di)	0.881	43.6	8.81
CD45RA(Er166Di)	0.881	40.3	5.28
MKI67(Yb172Di)	0.881	38.2	5.94
CD15(Sm149Di)	0.881	36.1	4.23
CD127(Pt195Di)	1.005	31.9	6.81
CD138(Er168Di)	1.444	28.5	6.08
CD141(Er170Di)	1.444	25.6	6.69
CD45RO(Yb173Di)	1.585	25.4	7.14
SMA(Pr141Di)	1.546	25.4	7.93
CD68(Tb159Di)	1.487	24.7	7.22
CD103(Gd158Di)	1.444	23.7	5.62
FoxP3(Dy162Di)	1.476	22.5	6.22
TTF1orCD16(Pt194Di)	1.630	20.7	4.97
LAMP3(Dy163Di)	1.894	16.9	7.18
TIM3(Nd143Di)	1.722	16.9	6.87
CD28(Yb174Di)	1.818	14.9	6.16
CD1c(Sm154Di)	2.447	8.2	6.22
CD14(Lu175Di)	2.422	6.6	6.65
CD38(Pt198Di)	1.919	6.2	4.99
HLADRB(Yb171Di)	3.835	3.8	8.43